19 To go beyond the two-dimensional metaphor of a balloon's surface and have a spherical three-dimensional model is easy mathematically but difficult to picture, even for professional mathematicians and physicists. You might be tempted to think of a solid, three-dimensional ball, like a bowling ball without the finger holes. This, however, isn't an acceptable shape. We want all points in the model to be on a completely equal footing, since we believe that every place in the universe is (on average) just like any other. But the bowling ball has all sorts of different points: some are on the outside surface, others are embedded in the interior, one is right in the center. Instead, just as the two-dimensional surface of a balloon surrounds a *three*-dimensional spherical region (containing the balloon's air), an acceptable round three-dimensional shape would need to surround a *four*-dimensional spherical region. So the three-dimensional spherical surface of a balloon in a four-dimensional space is an acceptable shape. But if that still leaves you groping for an image, do what just about all professionals do: stick to the easy-to-visualize lower-dimensional analogies. They capture almost all of the essential features. A bit further on, we consider three-dimensional flat space, as opposed to the round shape of a sphere, and that flat space can be visualized.

20 Depending on whether the rate of the universe's expansion is speeding up or slowing down over time, the light emitted from such galaxies may fight a battle that would have made Zeno proud: the light may stream toward us at light speed while the expansion of space makes the distance the light has yet to cover ever larger, preventing the light from ever reaching us. See notes section for details.10

21 Just as the video game screen gives a finite-sized version of flat space that has no edges or boundaries, there are finite-sized versions of the saddle shape that also have no edges or boundaries. I won't discuss this further, save to note that it implies that all three possible curvatures (positive, zero, negative) can be realized by finite-sized shapes without edges or boundaries. (In principle, then, a space-faring Magellan could carry out a cosmic version of his voyage in a universe whose curvature is given by any of the three possibilities.)

22 Today, matter in the universe is more abundant than radiation, so it's convenient to express the critical density in units most relevant for mass—grams per cubic meter. Note too that while 10^{-23} grams per